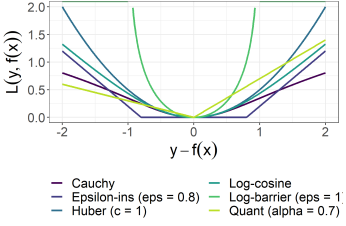
Introduction to Machine Learning

Advanced Risk Minimization
Advanced Regression Losses

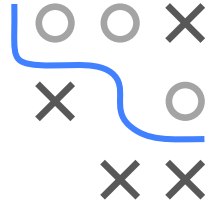


Learning goals

- Huber loss
- Log-cosh loss
- Cauchy loss
- ϵ-insensitive loss
- Quantile loss

ADVANCED LOSS FUNCTIONS

Wang et al. 2020

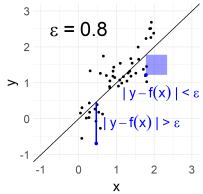


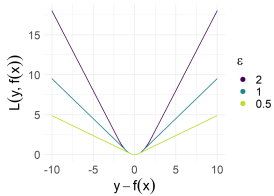
- Can use other losses for other posteriors
- Handle errors in a custom fashions
- Achieve properties like robustness
- Handle new predictive tasks

HUBER LOSS

$$L(y, f) = \begin{cases} \frac{1}{2}(y-f)^2 & \text{if } |y-f| \leq \epsilon \\ \epsilon|y-f| - \frac{1}{2}\epsilon^2 & \text{otherwise} \end{cases}, \quad \epsilon > 0$$

- Piece-wise combination of $L1/L2$ to have robustness/smoothness
- Analytic properties: convex, differentiable (once)





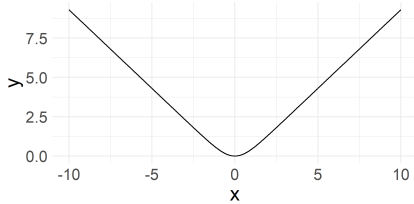
- No closed-form solution even for constant or linear model.
- Solution behaves like **trimmed mean**: a (conditional) mean of two (conditional) quantiles.

LOG-COSH LOSS

▸ R. A. Saleh and A. Saleh 2022

$$L(y, f) = \log\left(\cosh(|y - f|)\right) \quad \text{where } \cosh(x) := \frac{e^x + e^{-x}}{2}$$

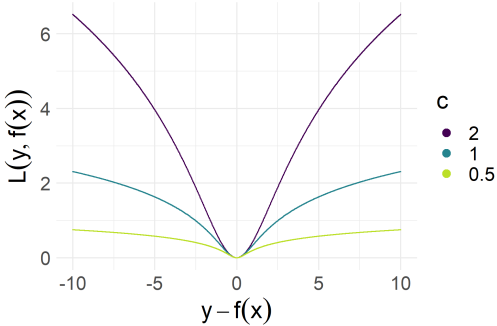
- Logarithm of the hyperbolic cosine of the residual.
- Approximately equal to $0.5(|y - f|)^2$ for small residuals and to $|y - f| - \log 2$ for large residuals, meaning it works a smoothed out $L1$ loss using $L2$ around the origin.
- Has all the advantages of Huber loss and is, moreover, twice differentiable everywhere.



CAUCHY LOSS

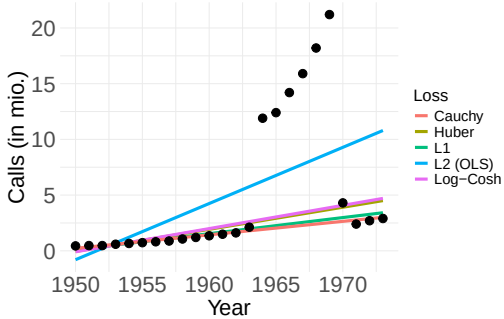
$$L(y, f) = \frac{c^2}{2} \log\left(1 + \left(\frac{|y - f|}{c}\right)^2\right), \quad c \in \mathbb{R}$$

- Particularly robust toward outliers (controllable via c).
- Analytic properties: differentiable, but not convex!



TELEPHONE DATA

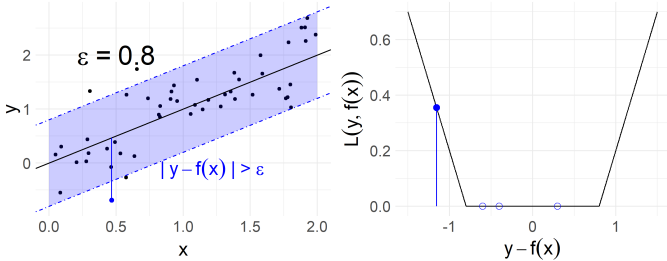
We now illustrate the effect of using robust loss functions. The telephone data set contains the number of calls (in 10mio units) made in Belgium between 1950 and 1973 ($n = 24$). Outliers are due to a change in measurement without re-calibration for 6 years.



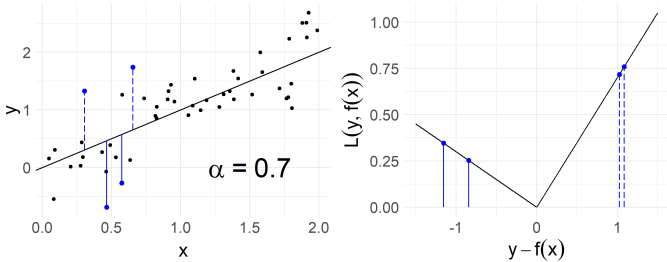
ϵ-INSENSITIVE LOSS

$$L(y, f) = \begin{cases} 0 & \text{if } |y - f| \leq \epsilon \\ |y - f| - \epsilon & \text{otherwise} \end{cases}, \quad \epsilon \in \mathbb{R}_+$$

- Modification of $L1$ loss, errors below ϵ accepted without penalty
- Used in SVM regression
- Properties: convex and not differentiable for $y - f \in \{-\epsilon, \epsilon\}$



QUANTILE LOSS / PINBALL LOSS

$$L(y, f) = \begin{cases} (1-\alpha)(f-y) & \text{if } y < f \\ \alpha(y-f) & \text{if } y \geq f \end{cases}, \quad \alpha \in (0, 1)$$

- Extension of $L1$ loss (equal to $L1$ for $\alpha = 0.5$).
- Weighs either positive or negative residuals more strongly
- $\alpha < 0.5$ ($\alpha > 0.5$) penalty to over-estimation (under-estimation)
- Risk minimizer is (conditional) α-quantile (median for $\alpha = 0.5$)